

Tensorial Observation Geometry in General Relativity

Coordinate covariance, frame/coframe duality, observation maps, and tensor descent

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Status. First tensor/GR reference document migrated to GO-QS v0.1. It corrects the tetrad reconstruction formula, declares the unit convention, states exact hypotheses for tensor descent, and separates coordinate, gauge, apparatus, and observation transformations. It does not modify Einstein's equations or claim a new theory of gravity.

Abstract

This note supplies a type-safe tensorial layer for Geometry of Observation. Four operations are separated: passive coordinate re-description, active diffeomorphism, local Lorentz frame gauge, and a possibly non-injective observation channel. Frame coefficients $e_a{}^\mu$ and coframe coefficients $e^a{}_\mu$ are distinguished; the covariant metric is reconstructed from the coframe, while the inverse metric is reconstructed from the frame. The default convention is signature $(-, +, +, +)$ and geometrized units $c = G = 1$, with the SI conversion stated explicitly. For a smooth observation map, the radical of a pulled-back pseudo-Riemannian form is computed exactly, showing why nontrivial kernel is sufficient but not necessary for degeneracy in Lorentzian signature. Smooth scalar descent and descent of covariant tensors along a surjective submersion are stated with their required fiber hypotheses. A local Lorentz gauge change is distinguished from a physical reorientation of an apparatus. A GO-QS quantity and protocol ledger closes the document.

Contents

1	Scope, status, and conventions	2
2	Four transformations that must not be conflated	2
2.1	Passive coordinate change	2
2.2	Active diffeomorphism	3
2.3	Local Lorentz frame gauge	3
2.4	Observation and reduction	3
3	Tensors and coordinate covariance	3
4	Metric, connection, curvature, and basepoint	4
5	Frame/coframe duality and the corrected tetrad formula	4
5.1	Example: spherical Minkowski coframe	5
6	Observers and transport	5
6.1	Vacuum is not flatness	6

7	Frame gauge versus physical apparatus orientation	6
8	The GO-QS tensor/GR observation package	6
9	Pullback geometry and its radical	7
9.1	No automatic Lorentzian distance	7
10	Descent and recoverability	7
11	Readouts on the frame bundle	8
12	Worked examples	8
12.1	Riemannian orthographic projection	8
12.2	Lorentzian null immersion	9
13	Quantity and Protocol Ledger	9
13.1	Coordinate-dimension rule	9
13.2	Minimal relativistic protocol	9
14	Correction log: v0.2 to v0.3	10
15	Claim audit	10
16	Conclusion	11

1 Scope, status, and conventions

Unless stated otherwise, manifolds are smooth, Hausdorff, and second countable. A spacetime (M, g) is a four-dimensional, time-oriented Lorentzian manifold with signature

$$\eta_{ab} = \text{diag}(-1, 1, 1, 1).$$

The curvature convention is

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

All relativistic formulae use the GO-QS unit context `GR_geometrized`: $c = G = 1$. Hence a proper-time parametrized four-velocity satisfies $g(u, u) = -1$. If proper time is measured in SI seconds and $u_{\text{SI}} = d\gamma/d\tau$, then

$$g(u_{\text{SI}}, u_{\text{SI}}) = -c^2.$$

Boundary 1.1 (Claim boundary). The tensor calculus, frame bundle, Fermi-Walker transport, and submersion results used below are standard. The contribution here is a corrected, typed interface to observation geometry. No new field equation, force, or gravitational degree of freedom is introduced.

2 Four transformations that must not be conflated

2.1 Passive coordinate change

Let $x : U \rightarrow \mathbb{R}^4$ and $x' : U' \rightarrow \mathbb{R}^4$ be charts. On the overlap,

$$x' \circ x^{-1} : x(U \cap U') \rightarrow x'(U \cap U')$$

is a diffeomorphism between coordinate images. It changes components, not the geometric point or tensor. With

$$J^\alpha{}_\mu = \frac{\partial x'^\alpha}{\partial x^\mu}, \quad (J^{-1})^\mu{}_\alpha = \frac{\partial x^\mu}{\partial x'^\alpha},$$

the basis and components transform contragrediently.

2.2 Active diffeomorphism

An active diffeomorphism $\varphi : M \rightarrow M$ maps fields by pullback or pushforward. Whether it represents a gauge redundancy, a physical symmetry, or a change of boundary data depends on the theory and boundary conditions. It is not identified with a passive chart change merely because both use Jacobians.

2.3 Local Lorentz frame gauge

For an orthonormal frame e_a , a local Lorentz transformation is

$$e'_a = \Lambda_a{}^b e_b, \quad \Lambda(x) \in \text{SO}^+(1, 3).$$

It changes the frame representation. If all frame components of physical fields are transformed consistently, gauge-invariant readouts do not change.

2.4 Observation and reduction

GO-QS assigns distinct signatures:

$$T_g : E \rightarrow E, \quad R_\lambda : E \rightarrow E_\lambda, \quad O_\lambda : E_\lambda \rightarrow Y_\lambda^0, \quad K_\lambda(dy \mid y^0).$$

T_g is invertible; R_λ and O_λ need not be. Information loss is therefore a property of a declared channel, not of covariance itself.

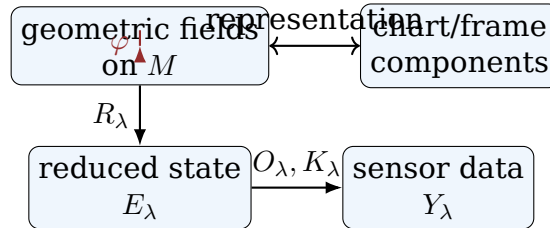


Figure 1: Representation changes preserve the geometric object; an observation pipeline can quotient or randomize it.

3 Tensors and coordinate covariance

Definition 3.1 (Tensor). At $p \in M$, a tensor of type (r, s) is a multilinear map

$$T : (T_p^* M)^r \times (T_p M)^s \rightarrow \mathbb{R}.$$

For $V = V^\mu \partial_\mu$ and $\omega = \omega_\mu dx^\mu$,

$$V'^\alpha = \frac{\partial x'^\alpha}{\partial x^\mu} V^\mu, \quad \omega'_\alpha = \frac{\partial x^\mu}{\partial x'^\alpha} \omega_\mu.$$

Consequently, the contraction $\omega_\mu V^\mu$ is coordinate independent. General tensor components receive one Jacobian per upper index and one inverse Jacobian per lower index.

Remark 3.2 (Coordinate passport). A component array is not a complete GO-QS quantity. It must carry the chart, basis, coordinate dimensions, index variance, and frame/gauge law. If coordinates have different physical dimensions, component dimensions are index dependent.

4 Metric, connection, curvature, and basepoint

The metric is

$$g = g_{\mu\nu} dx^\mu \otimes dx^\nu, \quad ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

Under a coordinate change,

$$g'_{\alpha\beta} = \frac{\partial x^\mu}{\partial x'^\alpha} \frac{\partial x^\nu}{\partial x'^\beta} g_{\mu\nu},$$

and the quadratic form ds^2 is unchanged. In Lorentzian signature, this quadratic form is not a metric distance function.

For $p \neq q$, the vector spaces $T_p M$ and $T_q M$ have no canonical identification. A connection supplies a transport rule. The Levi-Civita connection is characterized by

$$\nabla g = 0, \quad \text{Tor}_\nabla = 0,$$

and in coordinates

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}).$$

The Christoffel symbols are not tensor components. The curvature tensor is

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}.$$

The Ricci, scalar-curvature, and Einstein tensors are written

$$\text{Ric}_{\sigma\nu} = R^\rho_{\sigma\rho\nu}, \quad \text{Scal}_g = g^{\sigma\nu} \text{Ric}_{\sigma\nu},$$

$$\text{Ein}(g)_{\mu\nu} = \text{Ric}_{\mu\nu} - \frac{1}{2} \text{Scal}_g g_{\mu\nu}.$$

The notation $\text{Ein}(g)$ avoids a collision between the standard symbol $G_{\mu\nu}$ and the GO-QS symmetry-group symbol G . With geometrized stress-energy, Einstein's equation is

$$\text{Ein}(g)_{\mu\nu} = 8\pi T_{\mu\nu}.$$

At the SI boundary it is $\text{Ein}(g)_{\mu\nu} = (8\pi G_N/c^4) T_{\mu\nu}^{\text{SI}}$.

An affinely parametrized geodesic satisfies

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0.$$

Coordinate acceleration is not invariant; the covariant equation is.

5 Frame/coframe duality and the corrected tetrad formula

Definition 5.1 (Frame and coframe). A local frame and its dual coframe are

$$e_a = e_a^\mu \partial_\mu, \quad \vartheta^a = e^a_\mu dx^\mu,$$

with

$$e^a{}_\mu e_b{}^\mu = \delta^a_b, \quad e^a{}_\mu e_a{}^\nu = \delta_\mu{}^\nu.$$

The frame is orthonormal when

$$g_{\mu\nu} e_a{}^\mu e_b{}^\nu = \eta_{ab}.$$

Proposition 5.2 (Metric reconstruction). *For a Lorentz-orthonormal frame/coframe pair,*

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu, \quad g^{\mu\nu} = \eta^{ab} e_a{}^\mu e_b{}^\nu.$$

Proof. The abstract identity $g = \eta_{ab} \vartheta^a \otimes \vartheta^b$ gives the first component formula. Inverting the coframe matrix and the metric gives the second. Direct contraction with $e_c{}^\mu e_d{}^\nu$ returns η_{cd} . $\square$

Boundary 5.3 (Correction to v0.2). The expression $\eta_{ab} e_a{}^\mu e_b{}^\nu$ reconstructs $g^{\mu\nu}$, not $g_{\mu\nu}$. The covariant metric requires the dual coframe coefficients $e^a{}_\mu$.

5.1 Example: spherical Minkowski coframe

On a chart with $r > 0$ and $0 < \theta < \pi$,

$$g = -dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

in $c = 1$ units. An orthonormal coframe is

$$\vartheta^0 = dt, \quad \vartheta^1 = dr, \quad \vartheta^2 = r d\theta, \quad \vartheta^3 = r \sin \theta d\phi.$$

Then $g = \eta_{ab} \vartheta^a \otimes \vartheta^b$. Coordinate singularities at the polar axis are not curvature singularities.

6 Observers and transport

A future-directed observer is a timelike worldline

$$\gamma : I \rightarrow M, \quad u = \dot{\gamma}, \quad g(u, u) = -1.$$

Its acceleration is $a = \nabla_u u$, with $g(a, u) = 0$.

Definition 6.1 (Observer tetrad). An observer tetrad along γ is an orthonormal frame (e_0, e_1, e_2, e_3) with $e_0 = u$. The spatial axes span

$$u^\perp = \{X \in T_{\gamma(\tau)} M : g(X, u) = 0\}.$$

Definition 6.2 (Fermi-Walker transport). In the declared $c = 1$ convention, a vector field X along γ is Fermi-Walker transported when

$$\nabla_u X = \left(u \otimes a^\flat - a \otimes u^\flat \right) X.$$

For $a = 0$ this reduces to parallel transport.

In SI variables with $g(u_{\text{SI}}, u_{\text{SI}}) = -c^2$, the right-hand side is divided by c^2 . This conversion boundary is mandatory.

Proposition 6.3 (No canonical spatial triad). *At a point p , after fixing a unit future timelike vector u , the set of oriented orthonormal bases of $u^\perp$ is a torsor for $\text{SO}(3)$. Hence g and u do not select a preferred spatial triad.*

Proof. $u^\perp$ is a three-dimensional Euclidean vector space. $\text{SO}(3)$ acts freely and transitively on its oriented orthonormal bases. $\square$

6.1 Vacuum is not flatness

Einstein vacuum, $R_{\mu\nu} = 0$, does not imply $R^\rho{}_{\sigma\mu\nu} = 0$. For a Jacobi field ξ along a geodesic, the declared curvature convention gives

$$\nabla_u \nabla_u \xi = R(u, \xi)u.$$

Tidal relative acceleration can therefore provide a local relational signal even in vacuum.

7 Frame gauge versus physical apparatus orientation

The oriented, time-oriented orthonormal frame bundle is

$$\pi_{\mathcal{F}} : \mathcal{F}^+(M, g) \rightarrow M,$$

with structure group $\text{SO}^+(1, 3)$. A local Lorentz gauge transformation changes a representative frame. A physical apparatus, however, has material axes, couplings, and a response model. Rotating the apparatus relative to an external field changes its physical configuration; it is not erased by relabelling components.

Model 7.1 (Gauge-covariant sensor response). Let s denote physical fields and F a frame representative. A well-defined scalar readout $\mathcal{R}$ satisfies

$$\mathcal{R}(F\Lambda, \rho(\Lambda)^{-1}s) = \mathcal{R}(F, s)$$

for simultaneous passive frame change. An active apparatus rotation A acts on the apparatus state alone and may produce $\mathcal{R}(A \cdot F, s) \neq \mathcal{R}(F, s)$.

Boundary 7.2. The phrase “frame-dependent readout” is ambiguous unless it states whether the frame is a gauge representative or a material apparatus orientation. GO-QS stores these in different protocol fields.

8 The GO-QS tensor/GR observation package

For a relativistic application, use

$$\mathfrak{O}_{\text{GR}} = \left(E, \Phi, \xi, \Lambda, \{E_\lambda, Y_\lambda^0, Y_\lambda, R_\lambda, O_\lambda, K_\lambda, \hat{q}_\lambda\}_{\lambda \in \Lambda}, G_{\text{diff}}, G_{\text{LL}}, \mathcal{Q}, \mathcal{D} \right).$$

Here E is a state or field configuration space, not spacetime alone. G_{diff} and G_{LL} record their actions separately. A local protocol minimally contains

$$\lambda = (\gamma, u, \mathcal{E}_{\text{app}}, \varepsilon_x, \delta\tau, B_\nu, \Sigma_\eta, Q_{\text{quant}}, \mathcal{C}_{\text{coord}}, \mathcal{C}_{\text{LL}}, T_{\text{hor}}).$$

$\mathcal{E}_{\text{app}}$ is the material apparatus orientation; $\mathcal{C}_{\text{LL}}$ is a gauge convention.

9 Pullback geometry and its radical

Let $O : E \rightarrow B$ be smooth and let h be a nondegenerate pseudo-Riemannian metric on B . Define

$$(O^*h)_e(v, w) = h_{O(e)}(dO_e v, dO_e w).$$

The vertical space is $\mathcal{V}_e = \ker dO_e$.

Theorem 9.1 (Exact radical of a pullback form).

$$\text{rad}((O^*h)_e) = (dO_e)^{-1} \left((\text{im } dO_e)^{\perp_h} \right).$$

In particular, $\ker dO_e \subseteq \text{rad}((O^*h)_e)$.

Proof. A vector v belongs to the radical precisely when $h(dO_e v, dO_e w) = 0$ for every $w \in T_e E$. This is equivalent to $dO_e v$ being orthogonal to $\text{im } dO_e$, which gives the stated preimage. $\square$

Corollary 9.2. If h is Riemannian, then

$$\text{rad}((O^*h)_e) = \ker dO_e.$$

The same equality holds for pseudo-Riemannian h whenever O is a submersion at e .

Proof. For a positive-definite form, $\text{im } dO_e \cap (\text{im } dO_e)^{\perp_h} = \{0\}$. For a submersion, $\text{im } dO_e = T_{O(e)}B$, whose orthogonal complement is zero by nondegeneracy of h . $\square$

Remark 9.3. In Lorentzian signature, injectivity of dO_e does not guarantee nondegeneracy of O^*h : the image can be a degenerate null subspace. Thus “non-immersion implies degeneracy” is correct, but its converse fails.

9.1 No automatic Lorentzian distance

A Lorentzian metric supplies causal cones, proper time on timelike curves, and a time-separation function under suitable hypotheses. It does not supply a positive metric distance analogous to a Riemannian distance. Statistical residuals must use a positive-definite sensor covariance or a declared loss, not the indefinite form h as a norm.

10 Descent and recoverability

Theorem 10.1 (Smooth scalar descent). Let $O : E \rightarrow B$ be a surjective submersion and $q : E \rightarrow \mathbb{R}$ smooth. There is a unique smooth $\bar{q} : B \rightarrow \mathbb{R}$ satisfying

$$q = \bar{q} \circ O$$

if and only if q is constant on every fiber $O^{-1}(b)$.

Proof. Necessity is immediate. For sufficiency, define $\bar{q}(b) = q(e)$ for $e \in O^{-1}(b)$. A submersion admits smooth local sections $\sigma : U \rightarrow E$, and on U , $\bar{q} = q \circ \sigma$, proving smoothness. Surjectivity gives uniqueness. $\square$

Definition 10.2 (Horizontal covariant tensor). A covariant k -tensor T on E is horizontal for O if $T(v_1, \dots, v_k) = 0$ whenever at least one v_i is vertical.

Theorem 10.3 (Descent of covariant tensors). *Let $O : E \rightarrow B$ be a surjective submersion with connected fibers, and let T be a smooth covariant k -tensor on E . Then there is a unique covariant tensor $\bar{T}$ on B with*

$$T = O^*\bar{T}$$

if and only if:

1. T is horizontal; and
2. $\mathcal{L}_V T = 0$ for every smooth vertical vector field V .

Proof sketch. Pullbacks are horizontal and invariant under vertical flows. Conversely, horizontality makes the value of T depend only on projected tangent vectors. Vertical-flow invariance makes that value constant along each connected fiber. Local sections of the submersion define compatible local tensors on B , which glue uniquely. $\square$

Boundary 10.4 (Disconnected fibers and contravariant indices). If fibers are disconnected, vertical Lie invariance does not compare different components; explicit cross-component consistency is also required. The stated criterion is for covariant tensors. Descent of contravariant or mixed tensors requires a separate projectability condition and is not obtained by replacing indices formally.

11 Readouts on the frame bundle

Theorem 11.1 (Failure to descend to spacetime). *Let $\mathcal{R} : \mathcal{F}^+(M, g) \rightarrow Y$. If two frames F_1, F_2 over the same event satisfy*

$$\pi_{\mathcal{F}}(F_1) = \pi_{\mathcal{F}}(F_2), \quad \mathcal{R}(F_1) \neq \mathcal{R}(F_2),$$

then no map $\bar{\mathcal{R}} : M \rightarrow Y$ satisfies $\mathcal{R} = \bar{\mathcal{R}} \circ \pi_{\mathcal{F}}$.

Proof. Such a factorization would assign the same value $\bar{\mathcal{R}}(p)$ to both frames, contradicting the hypothesis. $\square$

This set-theoretic theorem does not decide whether the variation is a gauge artifact or a physical apparatus response. That decision requires the simultaneous transformation law in the protocol.

12 Worked examples

12.1 Riemannian orthographic projection

For

$$\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad \pi(x, y, z) = (x, y),$$

and $h = dx^2 + dy^2$,

$$\pi^*h = dx^2 + dy^2, \quad [\pi^*h] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Its radical equals $\ker d\pi = \text{span}\{\partial_z\}$.

12.2 Lorentzian null immersion

Let

$$i : \mathbb{R} \rightarrow \mathbb{R}^{1,1}, \quad i(s) = (s, s), \quad h = -dt^2 + dx^2.$$

di_s is injective, but

$$i^*h = -ds^2 + ds^2 = 0.$$

Thus an immersion can induce a degenerate pullback in indefinite signature. This is the counterexample excluded by an unjustified Riemannian intuition.

13 Quantity and Protocol Ledger

13.1 Coordinate-dimension rule

Store a coordinate passport $[x^\mu]$ for every chart. Since $[ds^2] = L^2$,

$$[g_{\mu\nu}] = \frac{L^2}{[x^\mu][x^\nu]}.$$

If all x^μ have length dimension, $g_{\mu\nu}$ is dimensionless. If $x^0 = t$ is measured in seconds while spatial coordinates are lengths, metric components have different dimensions and cannot be treated as an unannotated numerical matrix.

Object	Abstract dimension	GO-QS kind	Required metadata
x^μ	coordinate-specific	scalar/angle	chart and unit
g	$ds^2 : L^2$	covariant tensor	signature and coordinate passport
u	unitless in $c = 1$ normalization	vector	worldline and parameter
$a = \nabla_u u$	inverse length in geometrized parametrization	vector	unit context
$\Gamma^\rho_{\mu\nu}$	component-specific	connection coefficients	chart; not a tensor
$R^\rho_{\sigma\mu\nu}$	component-specific	tensor	curvature convention
Scal_g	L^{-2}	scalar	metric and sign convention
$e_a{}^\mu$	coordinate-specific	frame coefficients	frame, chart, Lorentz gauge
$e^a{}_\mu$	dual coordinate-specific	coframe coefficients	inverse-pair check
dO_e	output/input units	linear map	domain, codomain, rank threshold
O^*h	inherited from h	covariant tensor	signature and radical

13.2 Minimal relativistic protocol

Field	Required content
unit_context	SI or GR geometrized; explicit conversion boundary
signature	here $(-, +, +, +)$
curvature_convention	definition of $R(X, Y)Z$
coordinates	chart, ranges, units, singular sets
frame	frame and coframe matrices plus inverse-pair validation
lorentz_gauge	representative and transformation law
apparatus_orientation	material axes, distinct from gauge
observer	γ, u, a and transport law
observation	$R_\lambda, O_\lambda, K_\lambda$ with signatures
resolution	spatial, proper-time, spectral, and angular
uncertainty	covariance/posterior/risk and calibration
causal_domain	time orientation and admissible J^- region

14 Correction log: v0.2 to v0.3

1. Replaced the erroneous covariant metric reconstruction by the coframe formula and added the inverse-metric frame formula.
2. Declared signature, curvature convention, and GR_geometrized unit context.
3. Added the SI normalization $g(u_{\text{SI}}, u_{\text{SI}}) = -c^2$ and the $1/c^2$ Fermi-Walker conversion.
4. Separated passive coordinates, active diffeomorphisms, local Lorentz gauge, physical apparatus rotation, reduction, and observation.
5. Replaced the informal basic-tensor equivalence by a theorem for a surjective submersion with connected fibers.
6. Added the disconnected-fiber and mixed-tensor limitations.
7. Computed the exact radical of a pseudo-Riemannian pullback.
8. Added a null-immersion counterexample.
9. Explicitly denied a positive metric-distance interpretation for a Lorentzian form.
10. Added a coordinate-dimension passport and a complete GO-QS protocol ledger.

15 Claim audit

Established statements used:

1. tensor contractions are coordinate independent;
2. different tangent spaces require a transport rule;
3. frame and coframe reconstruct inverse and covariant metrics, respectively;
4. proper acceleration, Fermi-Walker rotation, and curvature are geometrically meaningful after conventions are fixed;
5. the radical formula characterizes local pullback degeneracy;

6. scalar and covariant-tensor descent require fiber compatibility.

Not claimed:

1. coordinate transformations are lossy observation channels;
2. every diffeomorphism is gauge under arbitrary boundary conditions;
3. an apparatus orientation is identical to a Lorentz gauge choice;
4. a Lorentzian metric is a positive distance;
5. every hidden tensor is recoverable;
6. this interface modifies general relativity.

16 Conclusion

The corrected tensorial interface is summarized by

covariance preserves representation-independent content;
a declared observation channel may quotient it.

The statement is now type safe. Coordinate and gauge changes act invertibly on representations; apparatus orientation belongs to the physical protocol; observation maps have explicit domains, codomains, and kernels; Lorentzian pullbacks are treated as indefinite bilinear forms; and tensor recovery is a descent problem with stated global hypotheses.

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